

## Introduction to Programming and Application Design

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### Mandatory Assignment # 2

#### **Deadline: Please check Exam Portal**

#### **Scenario:**

A local comic bookstore named ComicX sells comic books in city of Gotham. The store sells comic books which have 5 fixed themes named Batman, Joker, Cat, TheRiddler and Pinguin.

Each theme type books have fixed prices in respective order 22 USD, 13 USD, 11 USD, 15 USD, 19 USD). Bookstore owner Alfred wants to get a console app built for making his job easier. He wants to get a secure app so it gives access to its features only if correct username and password is given. Alfred wants to have following features/functionalities in his console job.

- a) Login using username and password (You can set a fixed username and password)
  - Print error if wrong username and/or password provided
  - On correct username and password display list of functions
- b) Following options should be shown to the user after successful login
  - Show number of Comic books Sold for each theme.
  - Show Total value of each theme type total books in stock.
  - Possibility to register sale of each comic book according to theme. //that means a function named OneBookSold(string themeType) that subtracts 1 from stock of books for that particular theme of comic books
  - A method that returns whether total stock is veryLow(less than 1000 books), Low(between 1000 and 1500 books), Normal (above 1500) or Over (above 5000). Total stock needs to be considered as **sum of all themes of comic books**.

#### **Tasks:**

Create a console application using Visual Studio and C# that performs following activity:

- When the app starts it prompts for a username:  
*"Please provide username to access the ComicX system"*
- Then prompt for a password  
*"Please provide password"*
- Please note there should be one standard username and password for the app. You select your own. If password and username are wrong. Please show following text and exit the app.  
*"You are not authorized to access this service"*
- On successful login a list of options should be presented:  
\*\*\*\*\* Here are your options \*\*\*\*\*  
*"Please select the action."*
  1. Show stock count for each theme of books
  2. Show total value of each theme type for all comic books in stock
  3. Register one comic book sold for a given theme
  4. Get stock status //veryLow, Low, Normal, Over
- At this point an input for 1 or 2 or 3 or 4 will be received from user and response will be shown

#### **Solution ideas**

There are multiple ways to solve the above assignment. One possible way could be as follows:

- Create a class named **ComicStock** and have properties for **ThemeName, Price, TotalInStockBatman, TotalInStockCat and so on..**
- Create a Processor class and add One method named Process() in that class.

- Create required methods in the classes
- In the Process Method start with writing code for creating object for Comic book Stock and fill it with any fake values.
- After that in the Process method write code for user interaction as stated above.
- Use methods wherever appropriate.
- In the Main method just call Process method.

**Important points to remember:**

- a) Please use proper variable names.
- b) Write three lines of what issues you had about writing this console app with respect to your understanding of programming so far.
- c) You can also make a different implementation than suggested above however you have to use classes and objects to pass this assignment.

Note: Please write the application code and submit in a pdf file. You can always print any file as pdf and then submit it.